

MARUAN BAKRI OTTONI

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EDUCATION

MSc in Computer Science, University of São Paulo (USP)

Ongoing

Research focus on computational neuroscience and random graph modeling of neural connections and social networks

Coursework: Graph Theory, Statistical Methods, Deep Learning, NLP, Parallel Computing

BSc in Electrical Engineering, University of Campinas (Unicamp), Brazil

2018 – 2022

GPA: 8.5/10

MSc/PhD level coursework: Information Theory, Stochastic Processes, Bayesian Inference, Machine Learning

Research Projects: Introduction to Lattice-based Cryptography(CNPQ Scholarship 2021-2022), Information Geometry for Machine Learning (CNPQ Scholarship 2022-2023)

PROFESSIONAL EXPERIENCE

Data Scientist II

September 2023 - Present

CloudWalk Inc.

São Paulo, Brazil

As part of an exploration team initiative, my job was to propose new state-of-the-art ideas to improve AI usage across the company. These include:

- Designed and deployed scalable monitoring tools to track the health of over 3000 ML features in our feature store, improving model reliability and observability.
- Built generative diffusion models for synthetic data generation in the context of tabular features for downstream tasks such as credit scoring and fraud detection, improving classification metrics by approximately 1%.
- Pretrained and finetuned foundational models for credit assignment using LLM-inspired architectures, enabling the deployment of credit scoring solutions that led to over R\$ 100 million in increased credit approval for clients.
- Developed predictive models for credit release performance, significantly accelerating the experimental phase and streamlining the deployment of new credit initiatives and models.
- Engineered risk assessment models for the credit portfolio to monitor and mitigate potential financial losses, ensuring the stability and health of the company's credit offerings.
- Led end-to-end development of experimental AI projects involving self-supervised learning, graph-based learning (GNNs) for fraud detection, and mechanistic interpretability for Large Language Models (LLMs).

Quant Analyst Summer Internship

January 2023 - March 2023

Goldman Sachs

São Paulo, Brazil

As part of a quant team I helped the development of a linear optimization software for asset inventory management across different entities of the bank

- Engineered algorithmic solutions that optimized asset allocation and significantly reduced redundant asset movements, improving operational efficiency.

Quant Analyst Internship

October 2022 - November 2022

UHedge Trading Solutions

Campinas, Brazil

- Conducted mathematical analysis of derivatives strategies to evaluate trading opportunities and risk-return profiles

RELEVANT SIDE PROJECTS

Logit Graph Developed a library for probabilistic graph generation via a logit-based edge process. Did also iterative benchmarks on current statistical graph models, achieving state of art performance for the method developed. [Lib](#)

Horror Movie Recommendation Built a FastAPI app that recommends horror movies from OMDb based on user prompts using a unified retrieval pipeline: Sentence-Transformers embeddings with blended priors (IMDb rating/votes, facet overlap, recency), optional cross-encoder reranking. Added UI, PostgreSQL, CI/CD, Payments integrations. [Code](#)

RAG Solution for Analyzing Auction Documents Developed a Retrieval-Augmented Generation (RAG) chatbot using Langchain and MongoDB, with a chat interface built in Flask. The bot enables interaction with potentially hundreds or thousands of documents across multiple auctions, allowing users to efficiently navigate and retrieve relevant information. [Code](#)

Graph Similarity via Representation Learning Contributed with ML methods for graph distance, implementing SOTA architectures and matching strong baselines. [Code](#)

SKILLS & LANGUAGES

Programming	Python, Matlab, C, Java
Tools	Google Cloud Platform, Docker, Git, Linux, Kubernetes, Spark, Airflow
ML/AI	PyTorch, TensorFlow, Scikit-learn, FSDP Training, Hugging Face
Data/Viz	SQL, Matplotlib, Seaborn, Plotly
Languages	Portuguese (Native), English (Fluent C2), French (Basic A2)